

Behavioral Institutionalism of the Agent-to-Agent (A2A) Economy: The A2A Institutional Dynamics Standard (A2A-IDS)

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ABSTRACT

The Agent-to-Agent (A2A) protocol specification v1.0.1 (Google, 2025; Linux Foundation) establishes syntactic interoperability across heterogeneous software agents via standardized discovery manifests (`/.well-known/agent-card.json`), JSON-RPC 2.0 message framing (`message/send`, `message/stream`), and a deterministic Task lifecycle (`SUBMITTED` → `WORKING` → `COMPLETED` / `FAILED` / `INPUT_REQUIRED` / `CANCELED`). Concurrently, the Model Context Protocol (MCP; Anthropic, 2024) standardizes agent perception filters through structured tool, prompt, and resource binding. However, syntactic conformance provides no instrument for assessing multi-agent institutional resilience, service-level veracity, or cascading fragility. This study formulates and empirically substantiates the *A2A Institutional Dynamics Standard (A2A-IDS)*. We project seven behavioral primitives onto protocol telemetry, formalize the A2A hub as an active informational field characterized by four physical parameters, and define *Institutional Anomie* (A_{hub}) as the Total Variation Distance between declared AgentCard capabilities and empirical log-derived execution clusters. In an empirical testbed of 500,000 tasks across 128 heterogeneous LLM agents, we validate econometric stationarity testing (ADF/KPSS) and Markov-chain spectral gap analysis ($\tau_{\text{rel}} = \frac{1}{1-|\lambda_2|}$), proving that sustained anomie ($A_{\text{hub}} > 0.30$) predicts cascading orchestration failures with an ROC-AUC of 0.912. We establish statistical justifications and ROC sensitivity analyses for all governance thresholds, develop tamper-evident cryptographic telemetry defenses against Goodhart's Law manipulation, and prove that the monitoring stack operates in $O(K)$ streaming time, adding less than 1.2% computational overhead.

Keywords: Agent-to-Agent (A2A) Protocol, Model Context Protocol (MCP), Digital Institutional Economics, Institutional Anomie, Total Variation Distance, Markov Transition Matrix, Spectral Gap, Goodhart's Law, Algorithmic Mechanism Design.

JEL Classifications: C73, D02, D82, L86, O33.

1. INTRODUCTION

1.1. Relevance and Problem Statement

The Agent-to-Agent (A2A) protocol specification v1.0.1 (Google, 2025; Linux Foundation, 2025) establishes syntactic interoperability among heterogeneous software agents through standardized discovery manifests (`/.well-known/agent-card.json`), JSON-RPC 2.0 message framing (`message/send`, `message/stream`), and a deterministic Task lifecycle (`SUBMITTED` → `WORKING` → `COMPLETED` / `FAILED` / `INPUT_REQUIRED` / `CANCELED`). However, this syntactic layer provides no instrument for assessing the evolutionary resilience of multi-agent environments. Three structural deficiencies motivate the present study.

First, the specification treats agents as opaque execution units. A hub can observe declared interfaces but cannot audit internal model weights, system prompts, or routing algorithms. Consequently, conformance to protocol grammar

does not entail conformance to declared service-level agreements (SLAs), generating an unmeasured gap between normative declarations and executed behavior.

Second, without a telemetry-based measurement apparatus, A2A hubs accumulate latent anomalies—*capability cloaking* (inflated `AgentSkill` declarations in the AgentCard), *cyclic task delegation* (Cycle Overflow across unmonitored agent chains), and *parasitic state retention* (Task Flooding via persistent `INPUT_REQUIRED` loops)—that degrade orchestration quality below detectable thresholds until cascading failure occurs.

Third, the Model Context Protocol (MCP; Anthropic, 2024) and emerging micropayment standards (HTTP 402 / x402) introduce additional interaction surfaces whose institutional stability remains unquantified in current literature on multi-agent systems (MAS), distributed AI governance, and digital institutional economics.

These deficiencies position the A2A hub not merely as a message router but as an institutional environment whose health requires continuous, mathematically grounded diagnostics—a gap this paper addresses.

1.2. Conceptual–Categorical Apparatus

The following constructs form the theoretical vocabulary of the study. Each is operationalized against observable A2A/MCP telemetry rather than inferred mental states:

- **(1) Informational Signal (S):** Any objectively logged event in the hub carrying source, carrier, form, and timestamp fields. In the A2A stack: AgentCard publications, `message/send` payloads, Task state transitions, and `x402` payment headers.
- **(2) Algorithmic Perception Filter (F):** The deterministic or stochastic configuration of a receiving agent that decodes S into an internal context: system prompt, context-window capacity, attached MCP tool set (`tools/list`), token budget, authentication schemas (`securitySchemes` per RFC 7515 / 8785), and LLM weight parameters.
- **(3) Stimulus (σ):** The structured token context assembled in the agent's prompt memory after S passes through F . σ is fully determined by $F(S)$ and contains no anthropomorphic affective component.
- **(4) Reaction (R):** An atomic, logged action or inaction: a Task state transition, emission of a content part (`TextPart`, `DataPart`, `FilePart`), invocation of an external MCP tool via `tools/call`, or connection timeout.
- **(5) Behavioral Pattern (Π):** A reproducible, autocorrelated sequence of reactions exhibited by one agent or a cohort under recurrent signal classes ($r > 0.70$ at lag 1, verified via Ljung-Box test, $p < 0.01$).
- **(6) Stationarity (Σ):** The statistical property that the distribution of reactions preserves its first two moments over successive observation windows W_t , verified via Augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests with $\text{TVD}(P_t, P_{t+1}) < 0.05$.
- **(7) Behavioral Spectrum (K_{spec}):** The full set of stable behavioral clusters ($K \geq 3$) emerging in a heterogeneous agent population in response to a single task class, consistent with Ashby's Law of Requisite Variety (1956).
- **(8) Institutional Anomie (A_{hub}):** The Total Variation Distance between the normative distribution Y (declared in AgentCards and protocol SLAs) and empirical behavioral distribution X (extracted from task logs via unsupervised clustering):

$$A_{\text{hub}} = \text{TVD}(Y, X) = \frac{1}{2} \sum_{i=1}^K |y_i - x_i| \quad (1)$$

- **(9) Agent-as-Proxy:** Every agent is an algorithmic executor of a human or organizational principal. The agent's filter encodes the principal's regulatory constraints, budget limits, and risk tolerances.
- **(10) Informational Field Parameters:** Four physically measurable quantities: signal density D_{hub} , propagation speed V_{prop} , access asymmetry A_{access} (Gini coefficient over message throughput), and Shannon entropy H_{hub} .

1.3. Formal Hypotheses

H1 (Anomie–Fragility Linkage): *If $A_{\text{hub}} > 0.30$ sustained over three consecutive observation windows, the probability of a cascading orchestration failure within the subsequent window exceeds 0.70, controlling for task volume.*

H2 (Spectral Compression under Asymmetry): *An increase in access asymmetry ($\text{Gini}_{\text{msg}} \rightarrow 1$) produces a statistically significant reduction in K_{spec} (toward monoculture), verified by a negative Spearman correlation ($\rho < -0.60$, $p < 0.01$) across hub operational logs.*

H3 (Markov Convergence Criterion): *The hub reaches a stationary institutional equilibrium if and only if the second-largest eigenvalue modulus of the $K \times K$ Markov transition matrix satisfies $|\lambda_2| < 0.85$, yielding a relaxation time $\tau_{\text{rel}} = \frac{1}{1-|\lambda_2|}$ bounded above by a policy-defined threshold.*

H4 (Autonomization Persistence): *Once a behavioral pattern achieves $K_{\text{aut}} > 0.80$, removal or modification of the originating hub signal does not dissolve the pattern within two subsequent windows, indicating self-sustaining institutional inertia.*

1.4. Methodology

The study employs a mixed-methods design combining formal econometric time-series modeling with empirical protocol trace analysis:

- (a) *Unsupervised behavioral clustering:* HDBSCAN (Campello et al., 2013) and Gaussian Mixture Models (Reynolds, 2009) extract natural reaction clusters from task-trajectory feature vectors (state sequences, p50/p95/p99 latencies, token consumption, error codes).
- (b) *Distributional divergence metrics:* Total Variation Distance (TVD) and Jensen–Shannon Divergence (Lin, 1991) quantify gaps between normative declarations (Y) and empirical execution (X).
- (c) *Time-series stationarity testing:* ADF (Said & Dickey, 1984) and KPSS (Kwiatkowski et al., 1992) tests confirm or reject unit roots on cluster-share time series $\{p_{i,t}\}$.
- (d) *Markov-chain spectral analysis:* Transition probability matrices $P = [p_{ij}]$ are evaluated for diagonal dominance ($p_{ii} > 0.85$) and spectral gap ($1 - |\lambda_2|$).
- (e) *Controlled perturbation experiments (A/B):* Hub configuration shifts evaluate pattern autonomization (K_{aut}) and filter feedback rates (Φ_{feed}).
- (f) *Informational-field parameterization:* Continuous telemetry stream computation of D_{hub} , V_{prop} , A_{access} , and H_{hub} .

2. PROJECTION OF SEVEN BEHAVIORAL PRIMITIVES ONTO THE A2A HUB ARCHITECTURAL STACK

The theoretical apparatus of behavioral institutionalism (North, 1990; Ostrom, 1990; Hodgson, 2006; Dyachkov, 2026a–g) is operationalized through seven sequential primitives that map exhaustively onto observable protocol fields:

Table 1. Operational Mapping of Behavioral Primitives to A2A / MCP Protocol Types

Primitive	Formal Specification	Protocol Implementation (A2A / MCP)	Server Telemetry Log Field
Signal (S)	Logged event with source, carrier, form, timestamp	<code>/.well-known/agent-card.json</code> ; <code>message/send</code> JSON-RPC; Task state changes; <code>x402</code> headers	<code>a2a.signal.id</code> , <code>event.timestamp_utc</code>
Filter (F)	Decoding configuration $\sigma = F(S)$	System prompt; context quota; attached MCP tools (<code>tools/list</code>); <code>securitySchemes</code> (RFC 7515/8785)	<code>mcp.tools_hash</code> , <code>agent.context_capacity</code>
Stimulus (σ)	Structured prompt context in token memory	Prompt memory context loaded prior to LLM inference	Inferred via prompt token length
Reaction (R)	Atomic logged machine action or timeout	Task transition (<code>WORKING</code> $\rightarrow$ <code>COMPLETED</code>); <code>DataPart / FilePart</code> emission; <code>tools/call</code> invocation	<code>task.state</code> , <code>mcp.tool_call</code> , <code>latency_ms</code>
Pattern (Π)	Autocorrelated sequence ($\rho > 0.70$)	Compliant execution chain; clarification loop; delegation cascade	<code>trajectory.ngram_hash</code> , <code>autocorr.rho</code>
Stationarity (Σ)	First two moments invariant across W_t	ADF $p < 0.05$ & KPSS $p > 0.05$; $\text{TVD}(P_t, P_{t+1}) < 0.05$	<code>stats.adf_pvalue</code> , <code>stats.tvd_delta</code>
Spectrum (K_{spec})	Cardinality of viable response modes	Distinct HDBSCAN reaction clusters with share $p_i \geq 0.05$	<code>hdbscan.cluster_count</code> , <code>k_spec.cardinality</code>

Source: Authors' formulation based on Google A2A v1.0.1 and Anthropic MCP (2024).

Algorithmic Perception Filter (F): Critically, F is heterogeneous across agents: two agents receiving an identical signal S will produce different internal stimuli if their filters differ in system instructions, attached MCP tools, or model weights. This heterogeneity is the proximate cause of behavioral spectral diversity and requires no anthropomorphic explanation.

Behavioral Spectrum (K_{spec}): Operationalizing Ashby's (1956) Law of Requisite Variety, spectral width is computed as:

$$K_{\text{spec}} = \sum_{i=1}^K \mathbb{I}(p_i \geq \varepsilon), \quad \varepsilon = 0.05 \quad (2)$$

3. THE A2A HUB AS AN INFORMATIONAL FIELD: FOUR MEASURABLE PARAMETERS

The A2A hub is reconceptualized from a passive router into an active informational field characterized by four physically measurable parameters:

Table 2. Mathematical Definition and Diagnostic Role of Informational Field Parameters

Field Parameter	Formula	Underlying Telemetry	Systemic Institutional Consequence
Signal Density (D_{hub})	$D_{\text{hub}} = \frac{1}{ T } \sum_{t \in T} \frac{M_t + \Delta S_t}{\Delta \tau}$	messages_per_task, tasks_per_agent_hour	Context saturation; forces agents into token evasion or silent timeouts.
Propagation Speed (V_{prop})	$V_{\text{prop}} = p50(\text{TTFM}^{-1})$	Time-to-First-Message (ms); speed G_{speed}	<i>Fait accompli</i> advantage: ultra-fast lightweight nodes capture routing graphs.
Access Asymmetry (A_{access})	$G_{\text{ini}_{\text{msg}}} = \frac{\sum_i \sum_j m_i - m_j }{2N \sum_i m_i}$	Outbound message/send share of top 1% nodes	Spectral compression: monopolistic broad-casters eliminate response diversity.
Field Entropy (H_{hub})	$H_{\text{hub}} = - \sum_{i=1}^K p_i \log_2 p_i$	Empirical share of outcome states in window W_t	Excess entropy indicates malformed payloads, cyclic delegation, and instability.

Formulation based on Shannon (1948) and Gini (1912).

4. FOUR COEXISTING INSTITUTIONAL REGIMES AS HYBRID CONFIGURATIONS

Empirical observation of enterprise systems confirms that multi-agent adoption does not proceed as a binary shift from human to autonomous execution. Rather, four interaction regimes operate simultaneously within the A2A infrastructure:

Table 3. Four-Regime Interaction Matrix with Protocol Primitives and Telemetry Standards

Regime	Sender → Receiver	Signal & Filter Architecture	Behavioral Spectrum	Diagnostic Governance Metric
H2H	Human → Human	Formal A2A Task; Filter: role authority, cognitive capacity	Accept, Delegate to Algorithmic Agent, Escalate, Reject, Request Info	human_response_latency, human_escalation_rate
H2A	Human → Agent	Natural intent parsed to Task; Filter: system prompt, MCP tools	Execute, Request Clarification (INPUT_REQUIRED), Policy Block, Sub-delegate	intent_to_task_conversion_rate, clarification_count
A2H	Agent → Human	INPUT_REQUIRED / AUTH_REQUIRED state; Filter: operator attention	Authorize, Ignore/Timeout, Re-route to Expert, Force Cancel (CANCELED)	escalation_response_time, escalation_abandonment_rate
A2A	Agent → Agent	JSON-RPC message/send, x402; Filter: opaque model weights	Complete (COMPLETED), Critical Fail (FAILED), Re-delegate, Timeout	task_completion_rate, delegation_depth, inter_agent_error_rate

Tripartite Coexistence: Three institutional arrangements persist in stationary equilibrium: (a) *Legacy Manual-Patched Workflows*, resolving algorithmic edge cases; (b) *Hybrid Human-in-the-Loop Automation*, maintaining financial authority and legal accountability (Dietvorst et al., 2015, 2018); and (c) *Fully Autonomous A2A Automation*, confined to high-frequency, low-liability circuits (e.g. x402 computational token settlements).

5. A2A ANOMIE: THE MEASURABLE GAP BETWEEN DECLARATION AND BEHAVIOR

In the A2A ecosystem, institutional anomie (Durkheim, 1897; Merton, 1938) is operationalized as the divergence between the machine-readable **Normative Layer** (Y) broadcast in AgentCards and the empirical **Behavioral Layer** (X)

observed in task traces.

For continuous governance optimization, the smooth Jensen–Shannon Divergence complements TVD:

$$\text{JSD}(Y \parallel X) = \frac{1}{2}D_{\text{KL}}(Y \parallel M) + \frac{1}{2}D_{\text{KL}}(X \parallel M), \quad M = \frac{1}{2}(Y + X) \quad (3)$$

5.4. Statistical Calibration and ROC Sensitivity Analysis of Diagnostic Thresholds

A frequent critique of threshold-based governance is the apparent arbitrariness of operational limits. To statistically calibrate the warning ($A_{\text{hub}} = 0.10$) and circuit-breaking ($A_{\text{hub}} = 0.30$) cutoffs, we conducted an empirical Receiver Operating Characteristic (ROC) sensitivity analysis across 50,000 sliding observation windows.

Table 4. Sensitivity, Specificity, and Youden's J Index Across Candidate Anomie Thresholds θ

Candidate Threshold θ		True Positive Rate (Sensitivity)	False Positive Rate (1 – Specificity)	Precision	F1-Score	Youden's J Index
$\theta = 0.10$ (Nominal Warning)		0.982	0.314	0.542	0.698	0.668
$\theta = 0.20$		0.941	0.176	0.702	0.804	0.765
$\theta = 0.30$ (Circuit Breaker)	(Circuit Breaker)	0.894	0.071	0.841	0.867	0.823
$\theta = 0.40$		0.682	0.024	0.912	0.780	0.658
$\theta = 0.50$		0.431	0.008	0.954	0.594	0.423

Evaluated on cascading failure detection ($\geq 25\%$ downstream failure in subsequent window).

Youden's J peaks decisively at $\theta = 0.30$ ($J = 0.823$), achieving an optimal balance between sensitivity (89.4%) and specificity (92.9%). Bounding nominal operations at $A_{\text{hub}} < 0.10$ and circuit breaking at $A_{\text{hub}} > 0.30$ is therefore statistically grounded in Bayes risk minimization.

5.5. Worked Protocol Example

An orchestrator declares $Y = (0.95, 0.05, 0.00)$ for SLA compliance, high latency, and critical failures. Empirical telemetry reveals $X = (0.35, 0.45, 0.20)$. The anomie index computes to:

$$A_{\text{hub}} = \frac{1}{2} \left(|0.95 - 0.35| + |0.05 - 0.45| + |0.00 - 0.20| \right) = \frac{1}{2} (0.60 + 0.40 + 0.20) = 0.60 \quad (4)$$

Because $0.60 \gg 0.30$, the node is quarantined immediately, preventing systemic propagation of routing failures.

5.6. Goodhart's Law and Adversarial Robustness of Telemetry

Per Goodhart's Law (1975), rational agents will attempt to manipulate telemetry once A_{hub} is codified as an enforcement trigger. A2A-IDS prevents metric falsification via three structural mechanisms:

- **Hub-Side Independent Wire-Tap Logging:** Telemetry is passively gathered at the TLS transport layer by the hub proxy, measuring physical arrival times, payload bytes, and HTTP return codes. Self-reported execution times are ignored.
- **Tamper-Evident Merkle Nonce Chains (RFC 7515 / 8785):** Each JSON-RPC exchange requires a sender-signed nonce verified against registered AgentCard public keys. State transitions are committed to a tamper-evident Merkle hash chain requiring reciprocal signatures from both interacting nodes.
- **Distributional Entropy & Benford Auditing:** Fabricated logs lack natural statistical variance. The hub monitors ∇H_{hub} and Benford distribution conformity in latency mantissas; synthetic smoothing triggers immediate quarantine.

6. EMPIRICAL VALIDATION AND EXPERIMENTAL RESULTS

To validate hypotheses **H1–H4**, we conducted a 14-day empirical evaluation comprising 500,000 multi-agent tasks executed across $N = 128$ autonomous LLM agents (Gemini 1.5, Claude 3.5, GPT-4o, Llama 3 70B) over 336 sliding observation windows ($\Delta t = 1$ hour).

Table 5. Empirical Evaluation of Cascading Failures under Varying Anomie Regimes (H1 Validation)

Anomie State	Observed Windows (N)	Observed Cascades	Empirical Cascade Probability	Downstream Error Rate
Nominal ($A_{\text{hub}} < 0.10$)	218	1	0.0046	1.2%
Spectral Expansion ($0.10 \leq A_{\text{hub}} \leq 0.30$)	76	4	0.0526	6.8%
Fragility Zone Sustained ($A_{\text{hub}} > 0.30$)	42	37	0.8810	38.4%

Binomial test vs. 0.70 null: $p = 0.0003$. ROC-AUC = 0.912.

Validation of H1: In 42 windows exhibiting sustained $A_{\text{hub}} > 0.30$, 37 resulted in cascading failure, yielding an empirical cascade probability of 0.881 (95% CI: [0.782, 0.980]), decisively confirming **H1** ($p < 0.001$).

Validation of H2: Broadcast asymmetry (Gini_{msg}) exhibited a strong negative rank correlation with spectral width K_{spec} ($\rho = -0.734$, $p = 0.0004$), confirming spectral compression toward monoculture.

Table 6. Markov Transition Matrix Spectral Parameters and Stationarity Tests Across Phases (H3 Validation)

Operating Regime	$ \lambda_2 $ Modulus	τ_{rel} (Windows)	ADF Stat (τ)	KPSS Stat	Institutional Stationarity Status
Equilibrium (Nominal)	0.762	4.20	-4.18 ($p = 0.0012$)	0.182 ($p > 0.10$)	Stationary digital institution certified
Schema Migration Shift	0.892	9.26	-2.41 ($p = 0.1420$)	0.541 ($p < 0.05$)	Metastable drift; active transition
Cycle Overflow Attack	0.984	62.50	-1.12 ($p = 0.7080$)	1.240 ($p < 0.01$)	Non-stationary; infinite cycle collapse

Validation of H3: Under nominal equilibrium, $|\lambda_2| = 0.762 < 0.85$, with ADF rejecting unit roots ($p = 0.0012$). During an induced Cycle Overflow attack, $|\lambda_2|$ rose to 0.984, relaxation time diverged to 62.5 windows, and stationarity failed, verifying **H3**.

Validation of H4: In controlled perturbation trials, agents maintained $K_{\text{aut}} = 0.838 \pm 0.042$ of established routing edges after prompt and tool modification, confirming institutional inertia.

7. HUB STABILITY: THE MEASUREMENT PROTOCOL

The measurement protocol operates through: (1) Sliding window discretization (P_t); (2) Inter-window drift quantification ($\text{TVD}(P_t, P_{t+1}) < 0.05$); (3) Econometric stationarity testing:

$$\Delta p_{i,t} = \alpha + \beta t + \gamma p_{i,t-1} + \sum_{j=1}^k \delta_j \Delta p_{i,t-j} + \varepsilon_t \quad (5)$$

and (4) Markov spectral gap relaxation tracking:

$$\tau_{\text{rel}} \approx \frac{1}{1 - |\lambda_2|} \quad (6)$$

8. MEASURABLE INSTITUTIONAL GAPS OF THE A2A HUB

The base A2A specification exhibits three institutional gaps: (a) an *economic vacuum* lacking escrow, resolved via HTTP 402 / `×402` micropayments; (b) an *opaque execution gap* where syntactic completion does not guarantee semantic validity, mitigated via zero-knowledge execution circuits; and (c) *human-in-the-loop safeguards*, where interrupted states (`INPUT_REQUIRED`, `AUTH_REQUIRED`) serve as embedded institutional circuit breakers.

9. HUB ARCHITECTURE AND COMPUTATIONAL COMPLEXITY ANALYSIS

The standard organizes the A2A hub into five functional layers: (1) Discovery & Identity (RFC 8785 JCS, RFC 7515 JWS); (2) Authorization & Escrow (CAAM, `×402`); (3) Task Orchestration (extended 8-state lifecycle: `SUBMITTED` → `WORKING` → `[DELEGATED | INPUT_REQUIRED | AUTH_REQUIRED]` → `[COMPLETED | FAILED | CANCELED]`); (4) Human Escalation; and (5) Institutional Telemetry.

Table 7. Specification of Five Novel Institutional Telemetry Metrics

Metric Name	Mathematical Formulation	Operational Interpretation	Governance Action Limit
<code>spectrum_width</code>	$K_{\text{spec}} = \sum_{i=1}^K \mathbb{I}(p_i \geq 0.05)$	Viable reaction cluster diversity	$K_{\text{spec}} > 3$ triggers schema tightening
<code>anomie_index</code>	$A_{\text{hub}} = \frac{1}{2} \sum_i y_i - x_i $	Normative vs. empirical SLA divergence	$A_{\text{hub}} > 0.30$ triggers quarantine
<code>stability_tvd</code>	$\text{TVD}(P_t, P_{t+1}) = \frac{1}{2} \sum_i p_{i,t} - p_{i,t+1} $	Inter-window behavioral drift	$\text{TVD} \geq 0.10$ signals restructuring
<code>autonomization_score</code>	$K_{\text{aut}} = \mathcal{E}_{\text{preserved}} / \mathcal{E}_{\text{baseline}} $	Pattern retention following signal perturbation	$K_{\text{aut}} > 0.80$ confirms institutional inertia
<code>filter_feedback_rate</code>	$\Phi_{\text{feed}} = N_{\text{distorted}}/N_{\text{total} \setminus \text{signals}}$	Distortion induced by accumulated error history	$\Phi_{\text{feed}} > 0.20$ requires memory reset

9.3. Computational Complexity and Streaming Scalability

A potential concern is whether continuous spectral monitoring imposes unacceptable computational overhead during high-frequency routing ($\Delta t = 15$ min). A2A-IDS achieves $O(K)$ streaming runtime through two architectural designs:

- **State Space Reduction** ($K \ll N$): Clustering and transition analysis operate on behavioral cluster archetypes ($K \in [3, 20]$), *not* raw agents ($N \approx 10^3 - 10^5$). Matrix operations on $K \times K$ take < 0.15 ms on a single core.
- **Online Power Iteration**: The second eigenvalue λ_2 is updated incrementally per transition via deflation against the stationary vector π :

$$v^{(m+1)} = \frac{(P - \mathbf{1}\pi^T)v^{(m)}}{\|(P - \mathbf{1}\pi^T)v^{(m)}\|} \quad (7)$$

Benchmark profiling across our 500k-task testbed demonstrated $< 1.2\%$ CPU overhead and < 65 MB RAM usage.

10. CONCLUSIONS AND SYNTHESIS OF FINDINGS

The A2A Institutional Dynamics Standard (A2A-IDS) transforms the A2A hub from a syntactic message switch into an active, self-diagnosing institutional environment. By parameterizing the hub informational field, formalizing

institutional anomie (A_{hub}) via Total Variation Distance, providing tamper-evident telemetry defenses against Goodhart's Law, and proving $O(K)$ streaming scalability, hub operators acquire an objective mathematical instrument stack for detecting structural fragility and preventing catastrophic multi-agent cascade failures before they materialize.

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